

Takoyaki Mini House System: A Web-based POS and Inventory Management Solution

IT 312L System Integration and Architecture 2

Tibar, Dina Mae M.

Jeremiah Dela Cruz

Adviser

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INTRODUCTION

In the highly competitive landscape of small food businesses, streamlining operations and improving decision-making through technology has become a critical factor for sustainability and growth. One of the most operationally intensive phases in a food establishment is the point-of-sale and inventory management process, where manual systems often lead to inefficiencies, data redundancy, and inaccurate stock tracking. In response to these challenges, this study introduces the Takoyaki Mini House System: A Web-Based POS and Inventory Management Solution—a full-stack web application designed to digitize and automate the sales and inventory workflows of Takoyaki Mini House, a small-scale takoyaki food business.

The system integrates point-of-sale functionalities with real-time inventory management through a centralized web-based platform. By replacing traditional paper-based logs and manual stock counting with a secure, role-based system, the proposed solution aims to reduce human error, prevent inventory shrinkage, and enhance transaction speed at the counter. Real-time data processing ensures that inventory levels are instantly updated upon each sale, while built-in reporting and visualization features support the owner in monitoring sales trends, identifying top-performing items, and adjusting inventory and pricing strategies accordingly.

This thesis documents the design, development, and implementation of the Takoyaki Mini House System, focusing on its impact on operational efficiency, data security, and managerial decision-making. The system is developed using modern web technologies, with an emphasis on usability, reliability, and scalability so that it may serve as a model for similar small food businesses seeking affordable and practical digital transformation. Through this study, the importance of integrating POS and inventory management into a unified, web-based solution is highlighted as a strategic step toward improving service quality, financial control, and long-term business resilience.

Background of the Study

The food service industry has become increasingly dependent on technology to improve operational efficiency, customer satisfaction, and financial accuracy. Small food businesses, such as takoyaki stalls and snack kiosks, often rely on manual methods for recording sales and tracking inventory. While these traditional methods may initially appear cost-effective, they frequently result in human errors, inaccurate inventory records, delayed reporting, and poor transaction management.

According to Oracle Corporation, Point-of-Sale (POS) systems improve business efficiency by automating sales processing, inventory monitoring, and report generation. Likewise, IBM emphasizes that inventory management systems help organizations reduce waste, monitor stock movement, and improve decision-making through accurate data analysis.

Takoyaki Mini House currently experiences challenges associated with manual transaction handling and stock monitoring. Daily sales are computed manually, product inventory is recorded using handwritten logs, and report preparation consumes significant time and effort. These operational issues increase the possibility of inconsistencies and financial inaccuracies.

To address these concerns, the proponents developed the Takoyaki Mini House System: A Web-Based POS and Inventory Management Solution. The system integrates sales processing, inventory monitoring, user management, and business analytics into a centralized digital platform accessible through a web browser.

The proposed system aims to improve operational performance, minimize manual errors, and provide real-time business information that supports managerial decision-making.

Purpose of the Study

The primary purpose of this study is to design and develop a web-based Point-of-Sale and Inventory Management System for Takoyaki Mini House. The proposed system aims to automate essential business processes, including order processing, sales recording, inventory tracking, and report generation, to improve operational efficiency and accuracy.

Specifically, the study seeks to provide an efficient and user-friendly platform that enhances transaction accuracy, improves inventory control, and supports data-driven management

practices. By centralizing sales and inventory information in a digital system, the proponents intend to reduce manual errors, minimize delays in reporting, and assist the business in making timely and informed decisions.

Importance of the Project

This project converts raw transactional and inventory data into actionable business intelligence that improves operational efficiency, profitability, and customer experience. For business owners, it replaces guesswork with data-driven demand forecasting and inventory optimization to reduce stockouts, overordering, and waste, while role-based access controls and audit logs provide a secure environment that enforces staff accountability and minimizes fraud or unauthorized changes. For managers, real-time dashboards and automated reports simplify decision-making and enable swift responses to sales trends, supplier performance, and seasonal demand. For customers, faster checkouts, accurate billing, and improved order accuracy reduce wait times and raise satisfaction—together strengthening operational resilience and fostering trust between the business and its customers.

Problems the System Aims to Solve

- Data redundancy and loss: replace fragile paper logs with a secure digital record to prevent physical damage, misplacement, and inconsistent entries; provide a single source of truth for transactions and inventory.
- Inventory shrinkage: link every inventory deduction to a verified transaction ID to improve traceability, discourage unrecorded removals, and simplify discrepancy investigations.
- Operational inefficiency: automate calculations for totals, change, and tax at the point of sale to shorten transaction time, increase customer throughput, and free staff to focus on service.
- Reporting delays: automatically collate daily, weekly, and monthly sales and inventory reports to eliminate hours of manual compilation and deliver timely insights for managers.

Objectives of the System

The main goal of this project is to design, develop, and implement a secure, full-stack, web-based Point-of-Sale (POS) and Inventory Management System for Takoyaki Mini House that leverages real-time data processing to optimize sales workflows, improve inventory accuracy, and support data-driven decision-making by the business owner and managerial staff.

General Objective

- To design, develop, and implement a secure, full-stack web-based POS and Inventory Management System for Takoyaki Mini House that uses real-time data processing to streamline sales operations, prevent stockouts and overordering, and provide timely business insights while maintaining data integrity and user accountability.

Specific Objectives

Security

- Implement a secure authentication module using Laravel's built-in encryption standards and session management to protect business data from unauthorized access.
- Enforce role-based access control (RBAC) so that only authorized users can perform sensitive operations such as voiding transactions, changing prices, or adjusting inventory.

Efficiency

- Create a "zero-reload" POS terminal using AJAX and the Fetch API to allow cashiers to add items, update quantities, and finalize transactions without page refreshes or interruptions.
- Minimize checkout time and reduce input errors to accommodate higher customer volume during peak hours.

Automation

- Develop an automated inventory trigger that decrements stock levels in real time whenever a sale is executed, ensuring that the inventory count always reflects actual usage.

- Optionally flag low-stock items and generate alerts so the owner knows when to reorder popular takoyaki variants or ingredients.

Intelligence

- Integrate data visualization tools (Chart.js) to display 14-day sales trends, category-wise revenue distribution, and top-selling items to help the owner identify patterns and adjust pricing or promotions.
- Provide simple, easy-to-read reports that support short-term planning, forecasting, and promotional decisions.

Accountability

- Maintain a detailed transaction log that records the exact timestamp, items sold, total amount, and the specific cashier responsible for each sale.
- Use audit trails to track voids, refunds, and discounts, which supports accountability and helps in resolving discrepancies or disputes.

Usability and Accessibility

- Design an intuitive web interface that is easy to learn for new staff, with clear buttons, visual cues, and minimal training requirements.
- Allow the system to be accessed from multiple devices within the store (e.g., tablets or laptops) so staff can operate from different counters without data loss.

Reliability and Data Integrity

- Ensure that transactions are stored consistently and that inventory records remain synchronized even during high-traffic periods.
- Implement backup and recovery mechanisms to protect against accidental data loss or system errors.

SCOPE AND LIMITATIONS OF THE PROJECT

What the System Can Do

- Multi-role access and dashboards: Provide separate, role-based dashboards for Admin (management) and Cashier (operations), with authentication and role-specific permissions (view-only vs. create/edit/delete).
- User management: Create, edit, deactivate, and audit user accounts; record basic activity logs (login times, role changes).
- Inventory lifecycle management: Add and edit products with images (supports WebP and common formats), SKU and barcode support, categorical assignment, pricing, stock levels, low-stock alerts, and expiration date tracking with configurable notifications.
- Product media and metadata: Store product images, optional descriptions and allergen tags, item weights or serving sizes (if provided), and toggles for active/inactive status.
- Dynamic POS terminal: Responsive, grid-based point-of-sale interface designed for touch screens, with category filters (e.g., Takoyaki, Drinks, Add-ons), live search, quick-add favorites, and custom modifiers (size, add-ons, quantity).
- Order handling and modifiers: Create and edit orders, apply discounts or price overrides (with permission checks), support voids/refunds, hold/open tabs, and combine or split receipts.
- Real-time financial settlement: Calculate totals, tax, discounts, and real-time change; validate payments for sufficiency before completing a transaction; show payment breakdowns (cash, coupon, etc.).
- Receipt and report generation: Generate instant thermal-style digital receipts (print dialog trigger) and downloadable executive PDF summaries, including daily sales, best-sellers, inventory valuation, and end-of-day cash reconciliation.

- Audit and reporting: Provide basic transaction logs, sales summaries by period, and exportable CSV/PDF reports for accounting.
- Responsive and cross-browser UI: Mobile- and tablet-friendly front end with layouts optimized for common modern browsers.
- Security basics: Encrypted password storage, session timeouts, and role-based access controls; basic input validation and server-side checks to prevent common mistakes.
- Extensibility: Designed to accept future modules (e.g., ingredient-level tracking, third-party payment gateways, hardware drivers) without rewriting core architecture.

Limitations and exclusions

- Connectivity and offline operation: The system is web-based and requires an internet connection or a local server for database synchronization. Offline-first or full offline operation (queuing transactions while disconnected and auto-syncing later) is not supported in the current version.
- Hardware integration: The application triggers the browser's print dialog for receipt printing but does not include proprietary drivers or guaranteed compatibility for specialized peripherals (pole displays, kitchen printers, cash drawers, barcode scanners). Integration may require additional middleware or vendor drivers.
- Payment methods: Optimized for in-person cash transactions ("Cash on Hand"). Built-in support for card payments, digital wallets, or third-party payment gateways is not included; these require separate integration and certification.
- Ingredient-level inventory: Tracks finished products (e.g., "6pcs Takoyaki") and product quantities only. It does not manage raw-ingredient inventories, recipe scaling, or per-ingredient consumption tracking (grams, liters).
- Multi-location and high-availability: Basic single-location installations are supported; advanced features like multi-store inventory syncing, distributed databases, automatic failover, or load-balanced hosting are out of scope.

- Regulatory and tax complexities: Provides simple tax calculation features, but advanced tax rules, multiple tax jurisdictions, or register-level tax compliance are not covered and may need localization or accountant review.
- Performance at scale: Designed for small-to-medium point-of-sale volumes. Very high throughput (thousands of concurrent POS terminals or extremely large catalogs) may require performance tuning, stronger infrastructure, or a different architecture.
- Data retention and backups: The baseline system offers export and manual backup options. Automated enterprise-grade backup, long-term archival policies, or immutable audit trails are not provided out-of-the-box.
- Accessibility and localization: Basic responsive design is included, but full accessibility compliance (WCAG 2.1 AA) and extensive localization (multiple languages, region-specific formats) may need additional work.
- Reporting depth: Executive PDF summaries and standard reports are available, but advanced analytics (trend forecasting, cohort analysis, complex KPI dashboards) are outside the initial scope.
- Customization and white-labeling: UI theming and minor config are supported; deep white-labeling or large-scale UI/UX customizations may require development resources.
- Security and compliance: Core security measures are implemented, but the product is not certified for specific regulatory standards (PCI-DSS for card data, HIPAA, or SOC2) in this release. Achieving those certifications requires additional controls and audits.
- Support and training: The offering includes basic documentation and onboarding guidance. Dedicated on-site training, 24/7 support, or premium SLAs are not part of the standard package.

Features Included in the System

1. Secure Role-Based Authentication System

- Multi-Level Access Control: Distinct interfaces for Admin (Full System Access) and Cashier (Operational Access).
- Encrypted Credentialing: Utilizes Laravel 12's BCrypt hashing for secure password storage.
- Session Persistence: Intelligent session management that redirects users to their appropriate starting point (Admin to Dashboard, Cashier to POS Terminal).
- Account Protection: Built-in safeguards to prevent the deletion of the primary Administrator account.

2. High-Performance Transaction Terminal (POS)

- AJAX "Zero-Reload" Engine: Processes additions to the cart and category filtering without refreshing the page, ensuring 100% uptime during busy hours.
- High-Density Product Grid: A compact, responsive grid (up to 6 columns) featuring high-visibility product cards with clear price highlights.
- Intelligent Inventory Guard: Real-time stock verification that prevents cashiers from adding "Out of Stock" items or exceeding current inventory limits in the cart.
- Live Search & Filtering: Dynamic search bar and category buttons that filter the menu instantly as the cashier types.
- Visual Indicators: Automated "Sold Out" overlays and "Low Stock" pulse alerts directly on the product cards.

3. Integrated Order & Cart Management

- Live Receipt Aesthetic: A streamlined "Active Cart" sidebar that mirrors the final receipt, showing a chronological list of items.
- Tactile Quantity Controls: Large, easy-to-tap "+" and "-" buttons for fast order adjustments.

- Automatic Financial Totals: Real-time calculation of Item Count, Subtotal, and Grand Total Payable.
- One-Click Clear: A safety-confirmed function to instantly wipe the cart for new customers.

4. Advanced Inventory & Product Lifecycle

- Full CRUD Suite: Complete Create, Read, Update, and Delete capabilities for the entire product catalog.
- Media Management System: Support for professional image uploads with compatibility for WebP, JPEG, and PNG formats.
- Automatic Storage Cleanup: A smart file manager that "unlinks" (deletes) old product images from the server when a product is updated or removed to save disk space.
- Expiration Tracking: Integrated native calendar pickers for monitoring product shelf-life and ensuring food safety.

5. Dynamic Category Control Center

- Relational Database Mapping: A dedicated module to manage product categories (e.g., "Classic Takoyaki," "Specialty Drinks") stored in a separate database table.
- POS Syncing: Any category added by the Admin automatically generates a new filter button in the POS Terminal.
- Safety Constraints: Prevents the deletion of categories that currently contain active products to maintain database integrity.

6. Executive Business Intelligence (Dashboard)

- Real-Time KPI Cards: Instant snapshots of Net Revenue, Order Volume, Active SKUs, and Restock Flags.
- Revenue Trend Visualization: A professional line chart (powered by Chart.js) showing the last 7 days of sales performance.
- Recent Activity Feed: A live log of the most recent transactions, including relative timestamps and cashier attribution.

7. Professional Settlement & Document Generation

- Floating Settlement Modal: A lightweight, non-intrusive payment window with a blurred backdrop for focused financial entry.
- Automated Change Calculator: Validates that the "Cash Received" is sufficient and calculates the exact "Return Change" instantly.
- Thermal-Style Receipt Generator: Produces a professional, digital "thermal paper" receipt after every sale, complete with transaction IDs and itemized breakdowns.
- One-Click PDF Export: A professional reporting tool that generates high-fidelity Executive Summary PDF reports for management meetings.

8. Sales History & Analytical Auditing

- Comprehensive Transaction Logs: A searchable database of every sale ever made, including exact dates, cashier names, and total amounts.
- Detailed Receipt Lookup: An "Eye Icon" feature that allows admins to reopen and view the full itemized receipt for any past transaction.
- Analytical Charts: Dedicated Bar Charts (14-day trends) and Doughnut Charts (Product Mix/Category Distribution) to visualize business growth.

9. Modern UI/UX Design (Elite POS Standard)

- Premium Typography: Built using the Plus Jakarta Sans typeface for maximum readability.
- Collapsible Sidebar: A space-saving navigation bar that can be hidden to maximize the workspace on small terminal screens.
- Persistent User State: Remembers the user's sidebar preference (Hidden/Unhidden) even after page refreshes.
- Slate & Orange Theme: A high-contrast, professional color palette optimized for retail environments.

System Description

Overview of the System

The system follows a Client-Server architecture. The backend is powered by Laravel 12, managing the "business logic" and database integrity, while the frontend provides a high-contrast, professional interface designed for high-glare kitchen environments.

Major Modules/Features

- **Administrative Back-Office:** A control panel for managing the "Product Master List," creating staff accounts, and auditing categories.
- **Transaction Terminal (POS):** The core operational module featuring a high-density product grid and an active cart sidebar.
- **Analytics Engine:** A reporting module that aggregates raw SQL data into visual graphs and itemized sales logs.
- **Storage Manager:** A system for handling product imagery, including automatic file cleanup when products are deleted.

User Roles and Functionalities

- **Admin:** Can perform CRUD on products/users/categories, view sensitive financial charts, and export business summaries.
- **Cashier:** Limited to the POS Terminal for order entry and the Sales History page for re-printing lost receipts.

Technologies Used

1. **Laravel 12 (PHP):** Chosen for its Eloquent ORM (Object-Relational Mapping), which ensures secure and efficient database queries.
2. **Tailwind CSS:** Used for the "Elite" UI design, providing a responsive and customizable layout without the overhead of heavy CSS files.
3. **MySQL:** Used for relational data storage, ensuring that transaction details are perfectly linked to specific products.
4. **JavaScript:** Utilized for the POS cart logic and asynchronous communication with the server.
5. **Chart.js:** A powerful JavaScript library used to render the sales trend and product mix visualizations.
6. **Plus Jakarta Sans:** A modern SaaS-standard typeface selected for maximum readability on digital terminals.

Use Case Diagram

The Use Case Diagram illustrates interactions between the system's two primary actors—Cashier and Administrator—and the Takoyaki Mini House System's major functionalities. Both actors must complete secure authentication before accessing any module. The Cashier performs operational tasks such as processing orders, accepting payments, printing receipts, and viewing sales history. The Administrator holds higher-level privileges, including inventory management, category management, user administration, analytics monitoring, and PDF report generation. Role-based access control ensures that each user accesses only the functionalities relevant to their responsibilities, providing a clear overview of the system's functional requirements and user interactions.

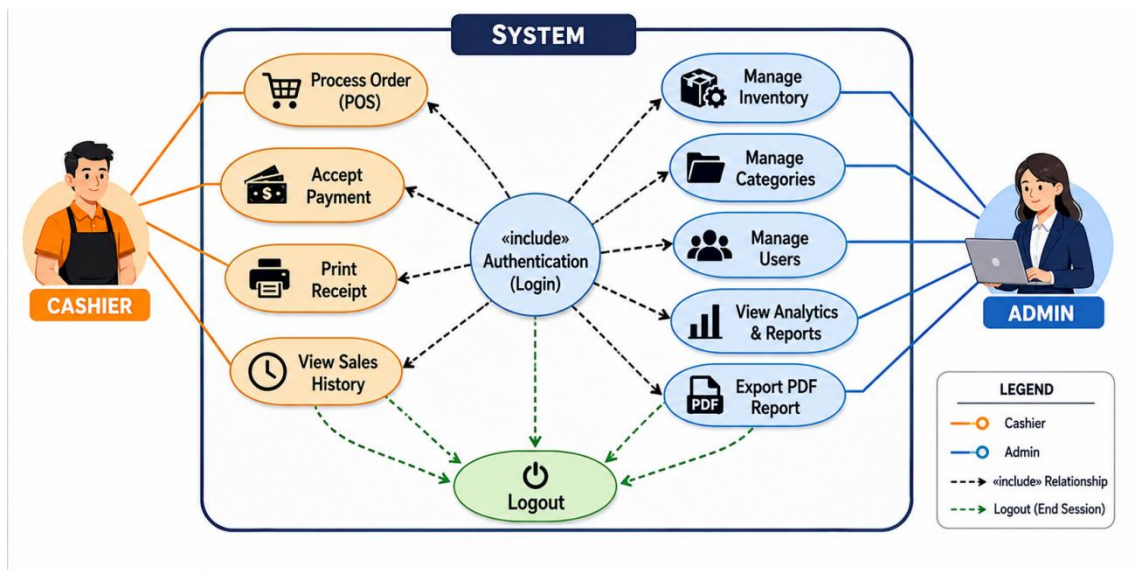


Figure 1: Use Case Diagram

Flowchart

The Flowchart presents the step-by-step POS transaction workflow. The process begins with user login authentication, followed by product selection and cart management. The cashier confirms the order and enters payment details. The system validates whether the payment is sufficient; if insufficient, it notifies the cashier and requests correction. Once payment is verified, the system processes the transaction, automatically deducts inventory quantities, generates a receipt, and stores the transaction record in the database. The flow concludes with successful transaction completion, visually representing the logical sequence of operations and highlighting the automation of sales and inventory functions.

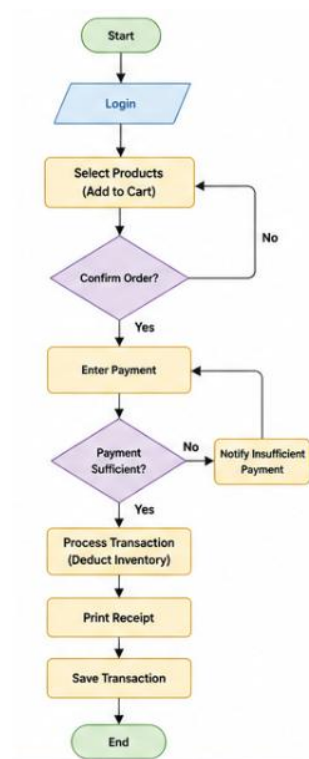


Figure 2: *Flowchart*

ER Diagram (Database Design)

The Entity Relationship (ER) Diagram represents the database structure of the Takoyaki Mini House System and shows relationships among five major entities: Users, Categories, Products, Transactions, and Transaction Details. The Users table stores account information for administrators and cashiers; the Categories table organizes products into groups; the Products table contains pricing, stock quantity, and expiration dates; the Transactions table records customer purchase information; and the Transaction Details table stores itemized sales records linked to each transaction. Relationships are established using primary and foreign keys to ensure data integrity and consistency, demonstrating how the relational database design supports efficient data storage, retrieval, and management.

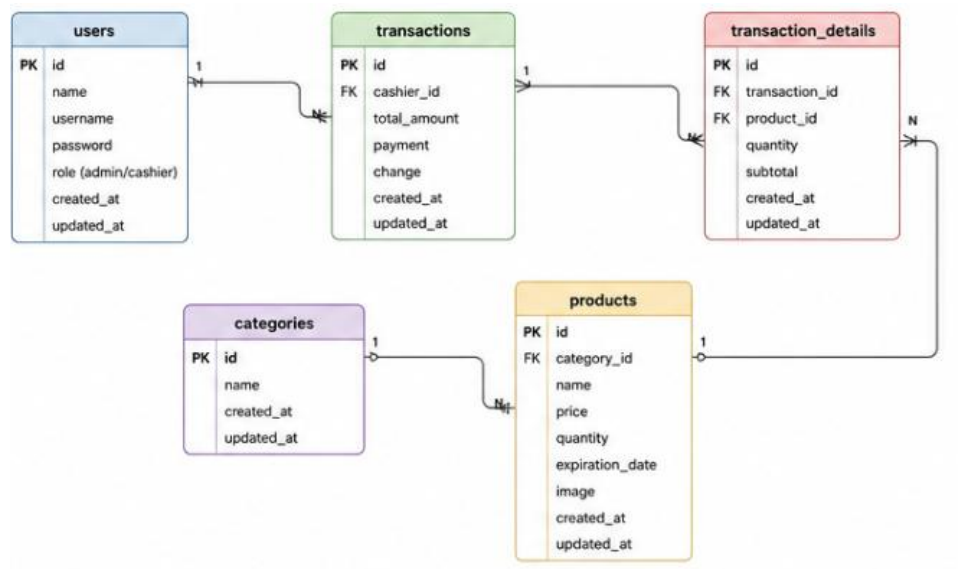


Figure 3: ER Diagram

System Architecture Diagram

The System Architecture Diagram illustrates the technical structure and communication flow of the Takoyaki Mini House System using a client-server model. Users access the system through a web browser interface built with React and Tailwind CSS. Client requests are sent via HTTP/HTTPS to a Laravel-based web server that processes business logic, authentication, inventory operations, and transaction management. The server communicates with a MySQL database for data storage and retrieval. Additional services such as PDF generation, file storage, and role-based authentication are integrated to enhance functionality and security. The diagram demonstrates the interaction among frontend, backend, and database components, highlighting the system's scalability and modularity.

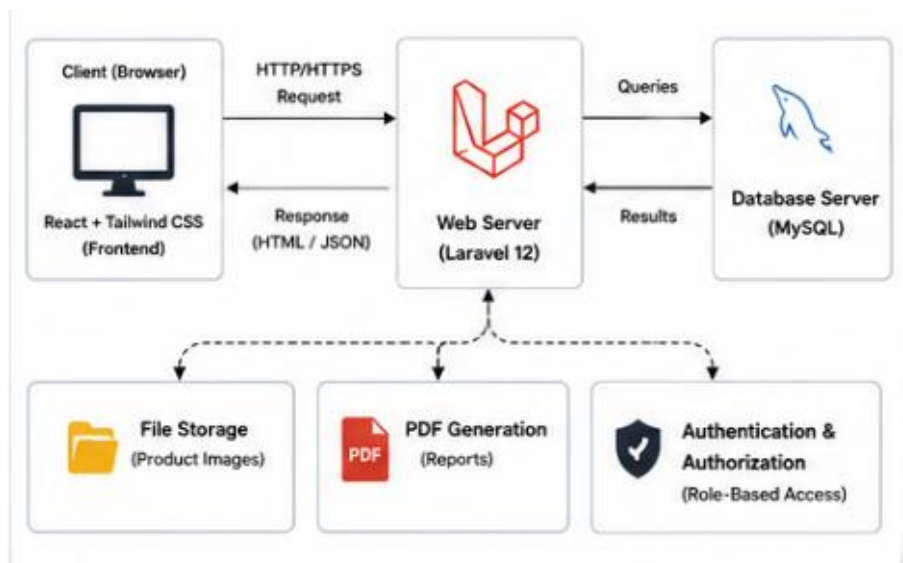


Figure 4: *System Design*

Database Structure

Tables and Fields

Table	Field	Data Type	Description
users	role	ENUM	Restricts access level (Admin/Cashier)
products	category_id	BIGINT (FK)	Links product to a specific category
products	price	DECIMAL(10,2)	Stores currency value with precision
transactions	total_amount	DECIMAL(10,2)	Final amount to be paid
transaction_details	quantity	INT	Number of items sold in that specific order

Table 1: Tables and Fields

Relationships

Relationship Type	Description
One-to-Many	One Category can have many Products
One-to-Many	One Transaction can have many Transaction Details (items)
Many-to-One	Many Transactions are processed by one User (Cashier)

Table 2: Relationships

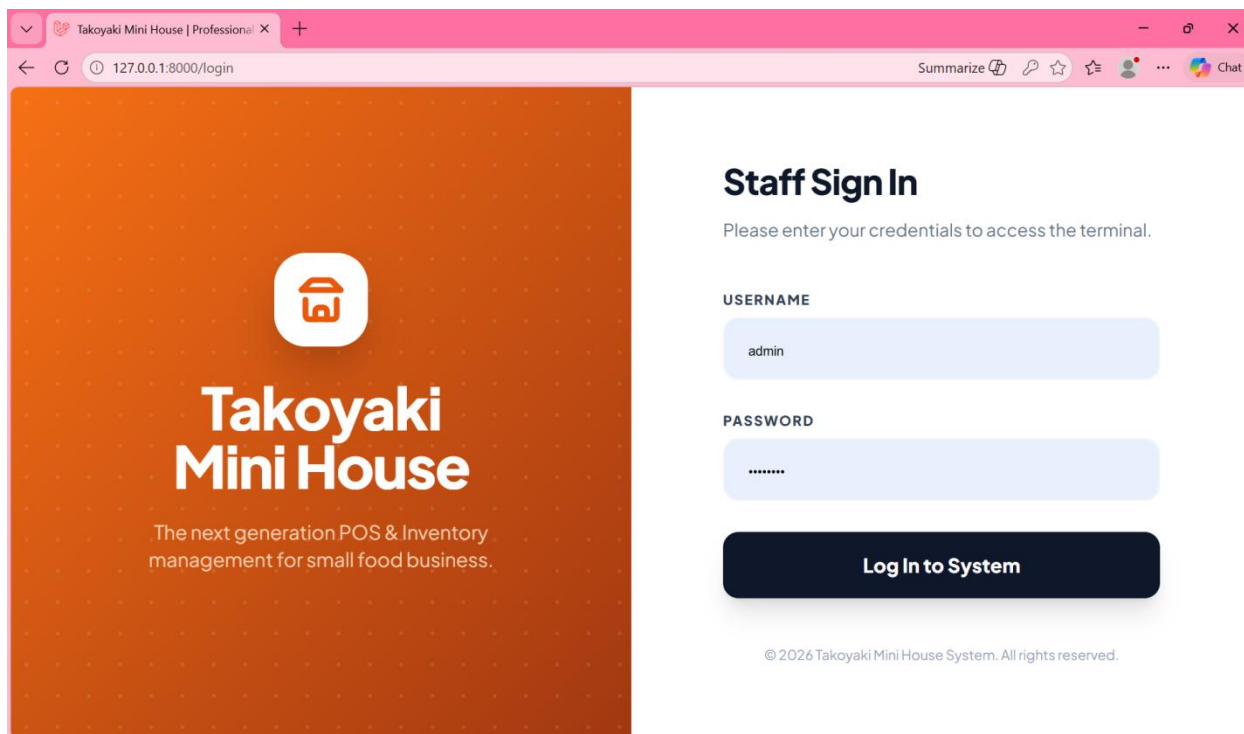


Figure 5: Login Page

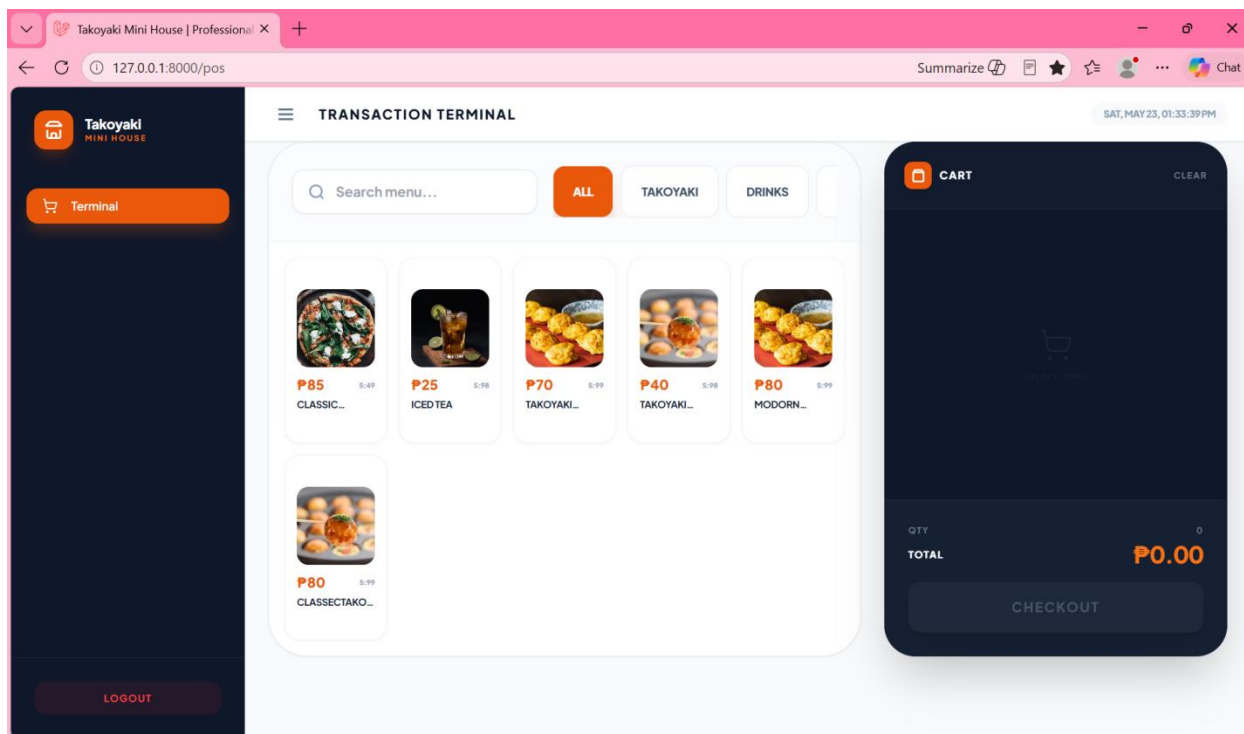


Figure 6: Cashier Terminal

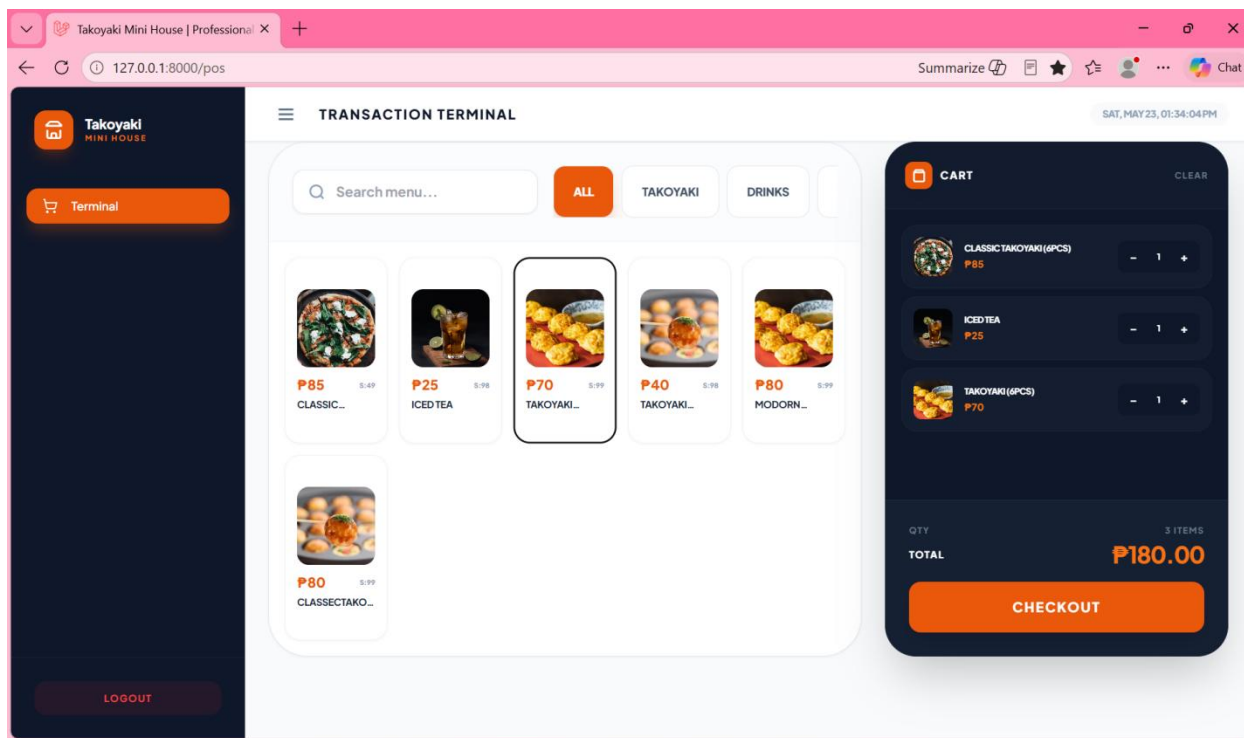


Figure 7: Add to Cart

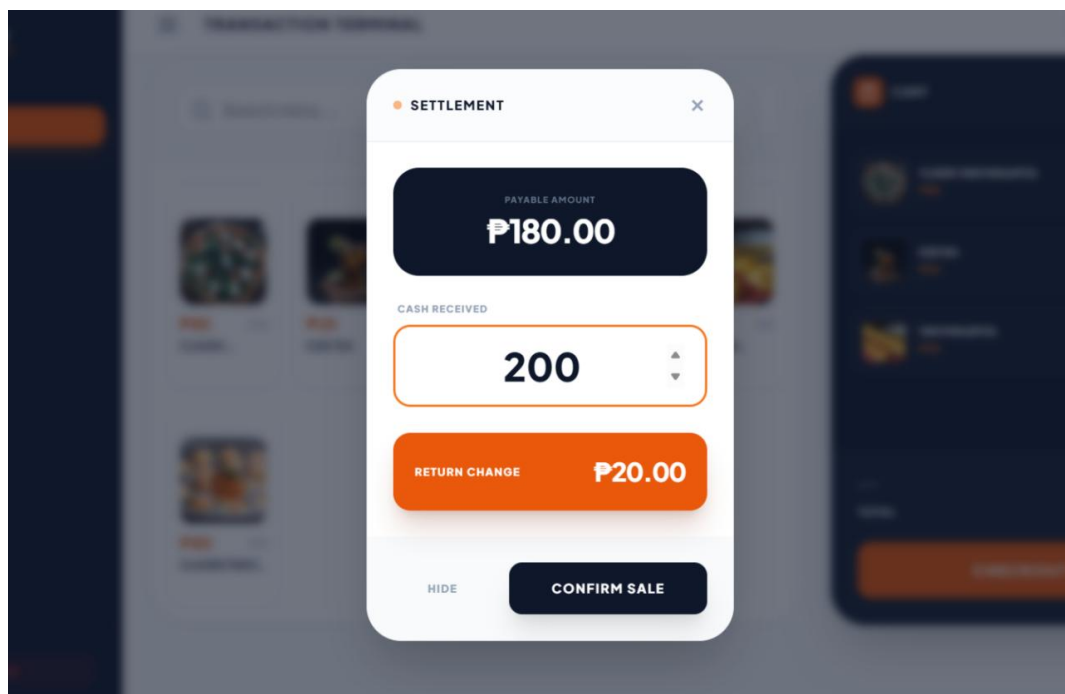


Figure 8: Input Money

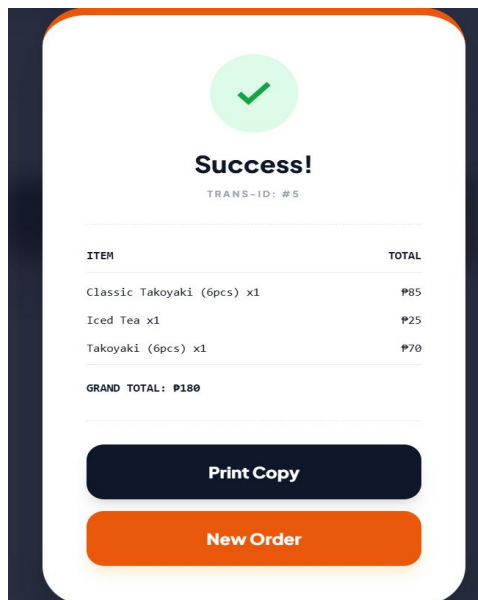


Figure 9: Receipt

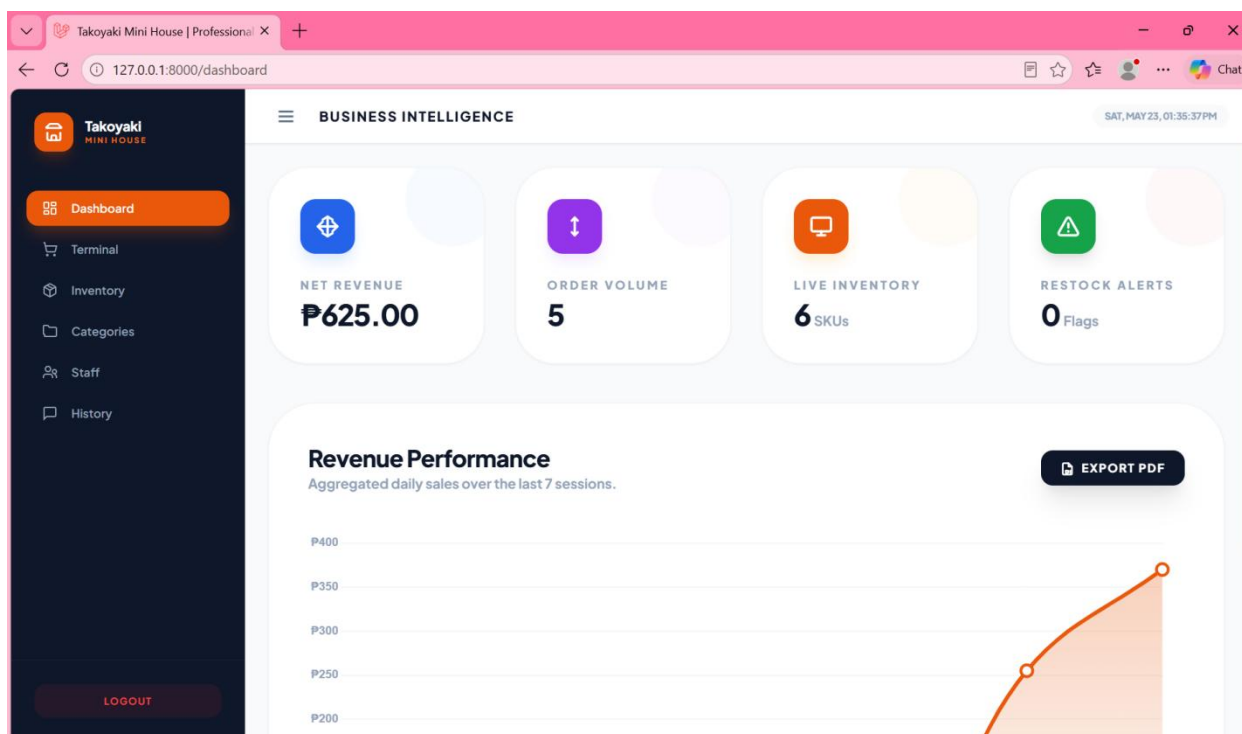


Figure 10: Admin Side

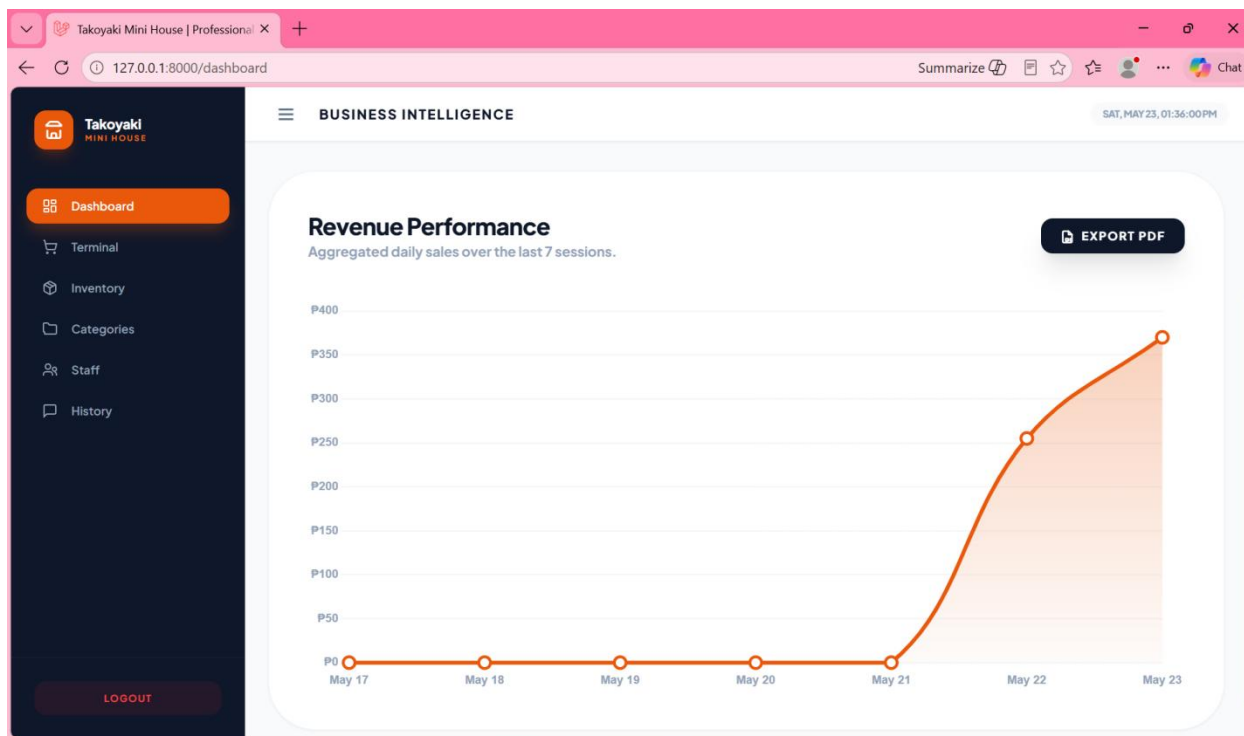


Figure 11: Admin Dashboard

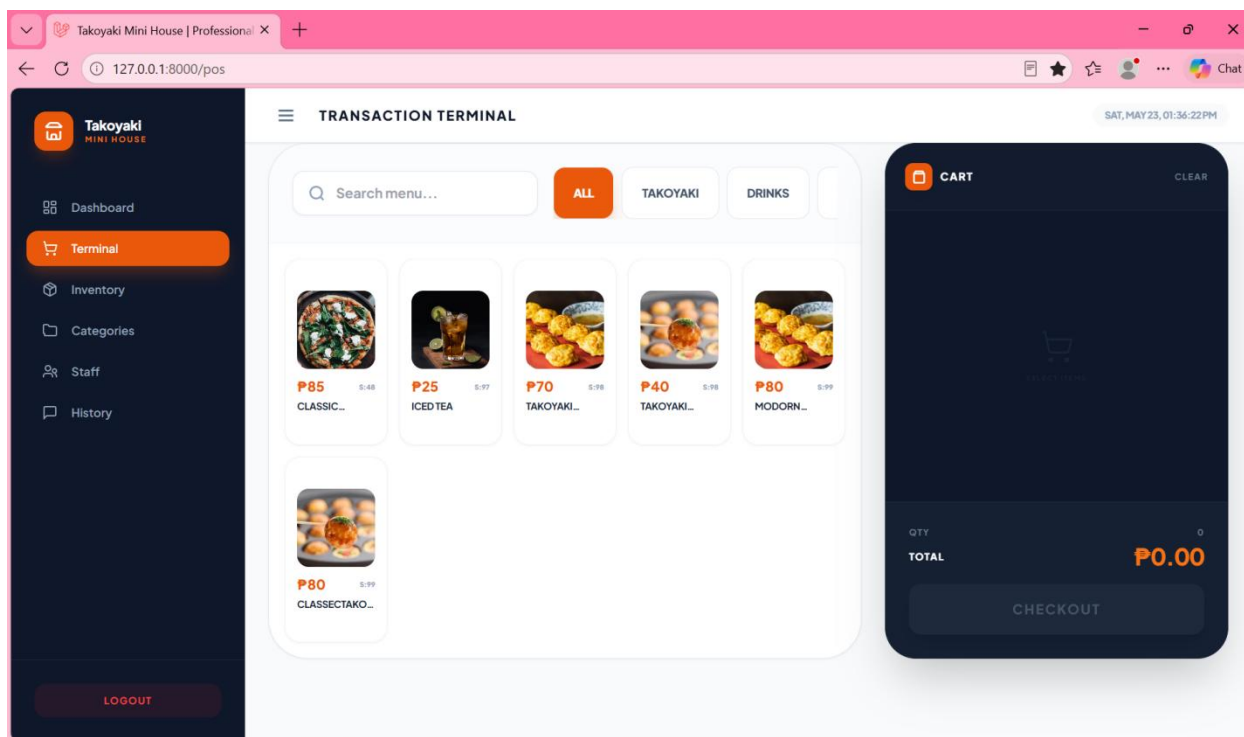


Figure 12: Admin Terminal/Menu

TAKOYAKI MINI HOUSE

Inventory Management
Monitor and manage your Takoyaki stock.

[+ Add Product](#)

PRODUCT INFO	CATEGORY	PRICE	QUANTITY	EXPIRATION	ACTION
Classic Takoyaki (6pcs) ID: #PROD-1	TAKOYAKI	₱85.00	48	2025-12-31	
Iced Tea ID: #PROD-2	DRINKS	₱25.00	97	2026-01-15	
Takoyaki (6pcs) ID: #PROD-3	ADD-ONS	₱70.00	98	2026-05-25	
Takoyaki (60pcs) ID: #PROD-4	TAKOYAKI	₱40.00	97	2026-05-25	
modern Takoyaki (60pcs) ID: #PROD-5	TAKOYAKI	₱80.00	98	2026-05-25	

Figure 13: Inventory

Add New Product

Product Image

Product Name

Takoyaki (60pcs)

Price (₱)

80

Quantity

100

Category

Takoyaki

Expiration Date

25/05/2026

Cancel

[Save Product to Inventory](#)

Figure 14: Add Items

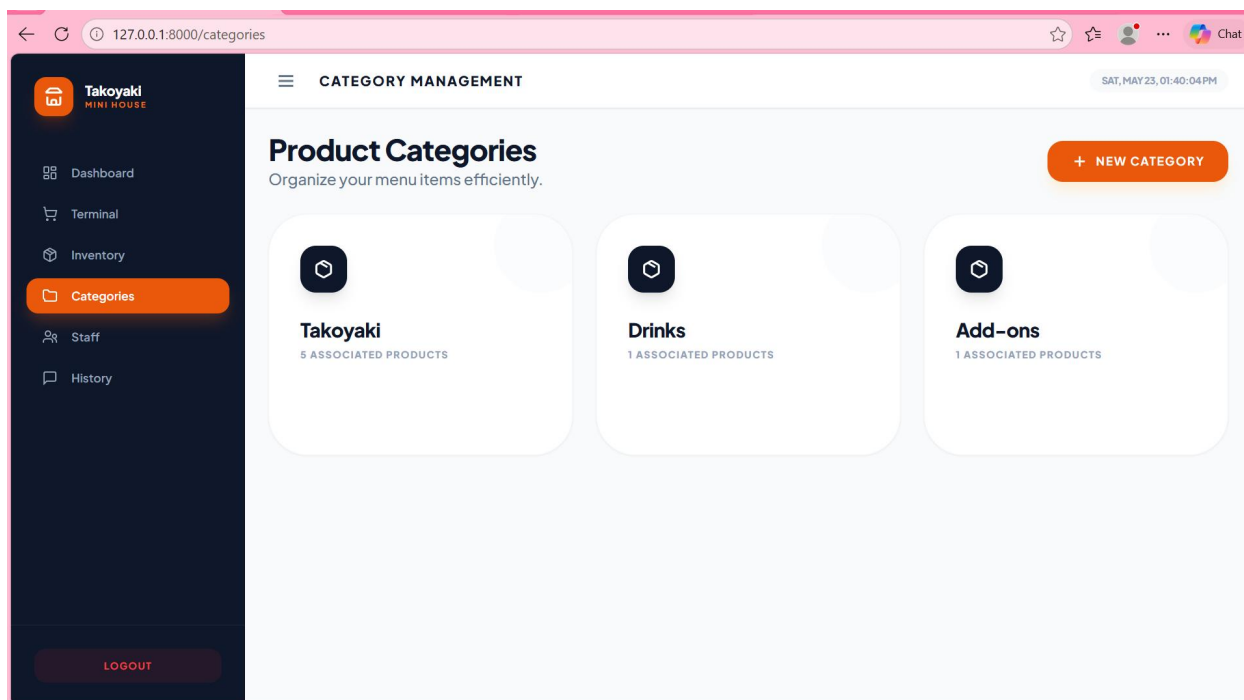


Figure 15: *Categories*

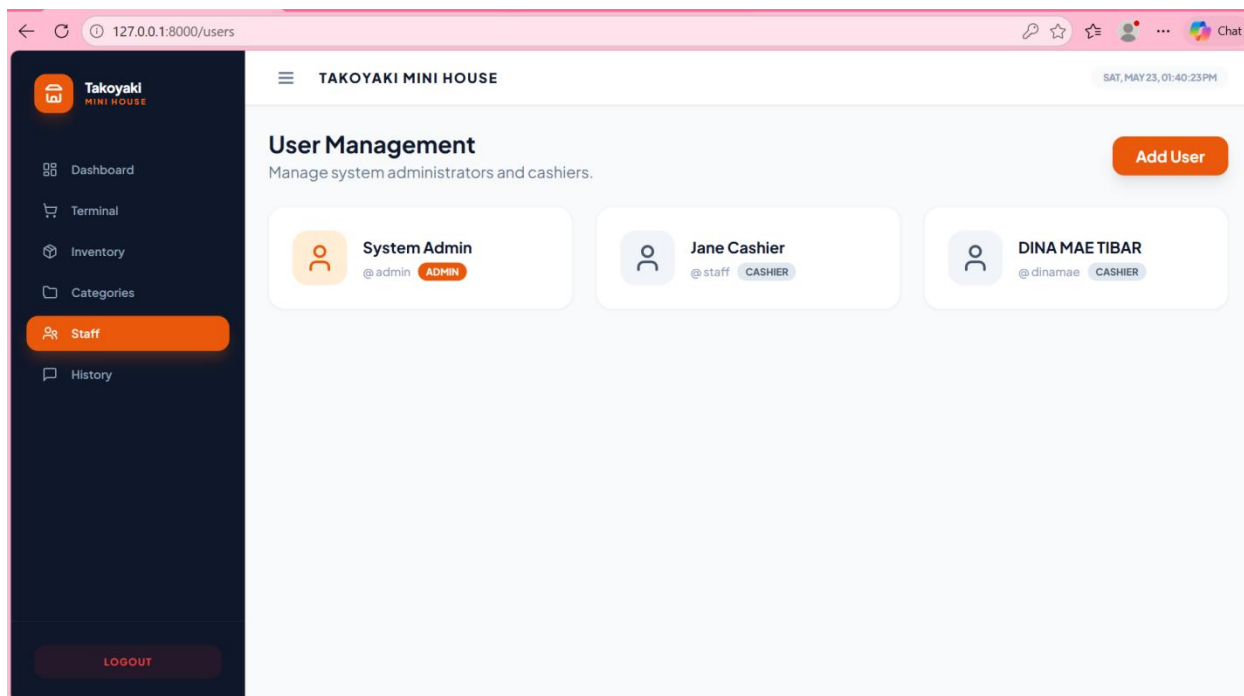


Figure 16: *Staffs*

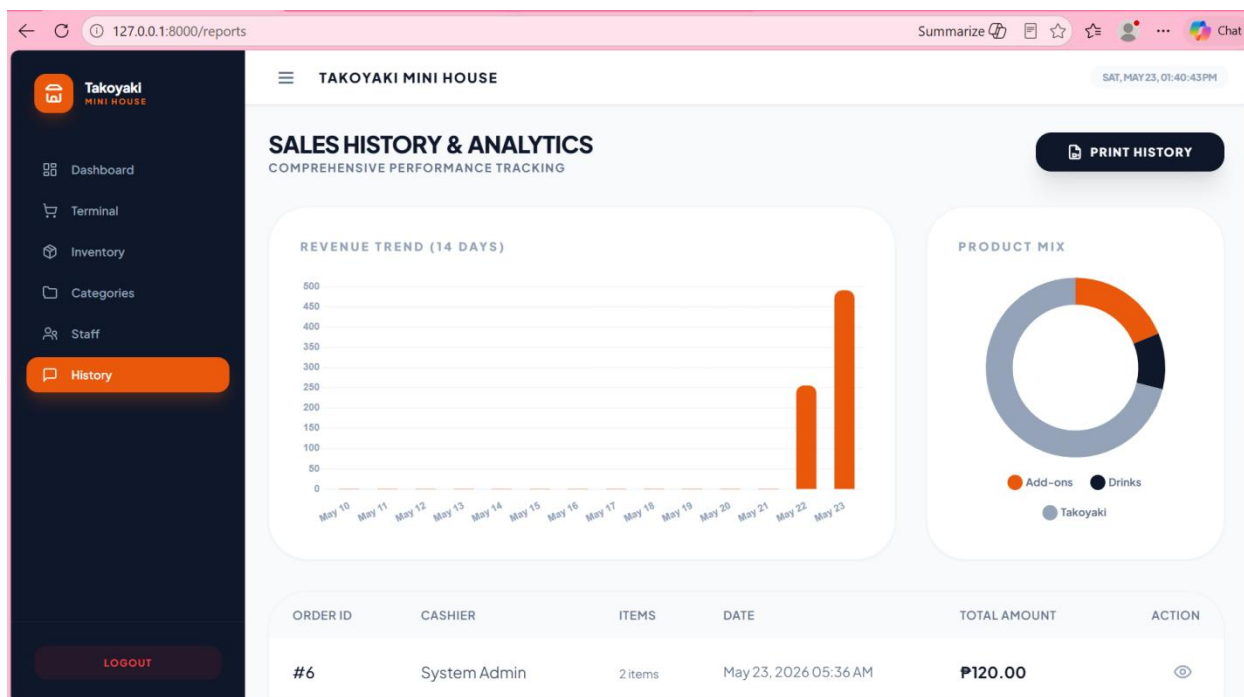


Figure 17: History

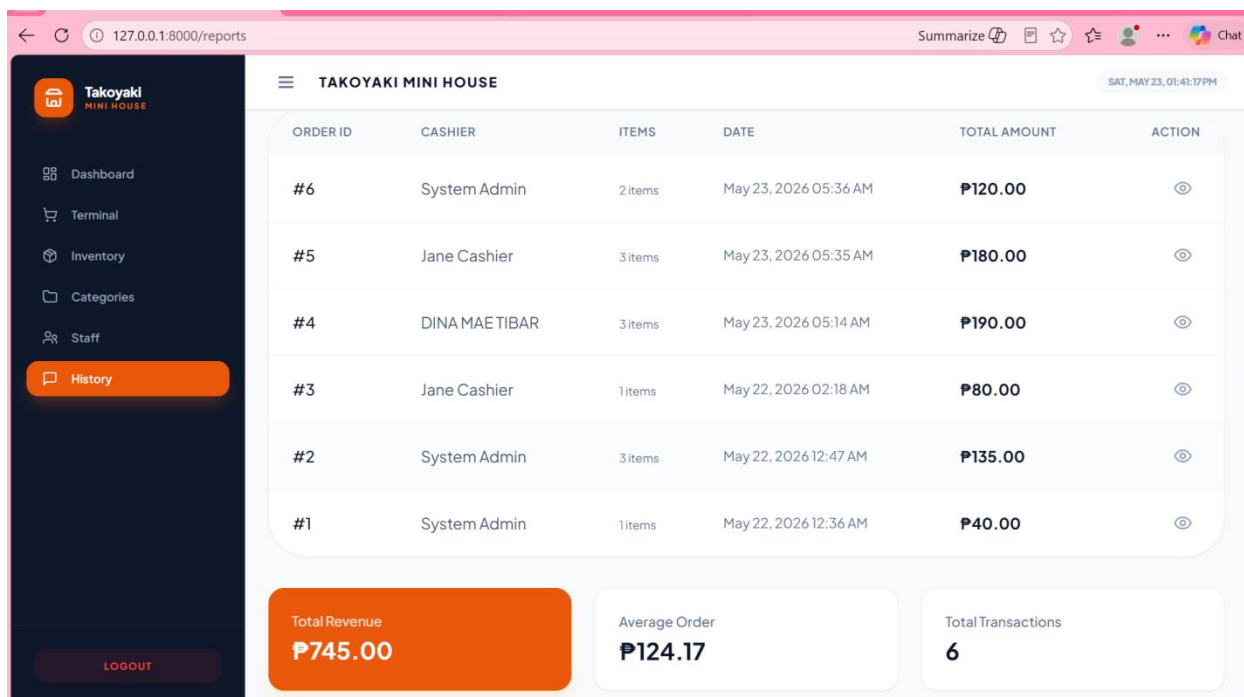


Figure 18: Records of Users

Implementation and Testing

Development was conducted in three phases. The Structural phase involved database migration and Model relationship mapping to establish the foundation of the system. The Functional phase focused on implementing the POS cart logic and Controller-based stock deduction to ensure accurate transaction processing and inventory management. The Visual phase applied the Tailwind CSS theme for a professional interface and integrated Chart.js for data visualization on the dashboard.

Testing procedures ensured system reliability and usability. Boundary Testing involved inputting 0 or negative values in the price and quantity fields to verify the system rejects invalid entries. Concurrency Testing simulated a sale while an admin updated the same product's price to confirm data consistency under simultaneous operations. Cross-Browser Testing verified functionality on Chrome, Edge, and Safari to ensure Floating Modals render correctly across all major browsers.

Two problems were encountered and resolved during development. SQL Foreign Key Errors occurred when deleting a product was blocked by existing sales records; this was solved by implementing a cleanup script in the destroy method and adding `onDelete('cascade')` to the migration. UI Clutter made the terminal too long with many products; this was solved by implementing a high-density 6-column grid with smaller, compact cards and truncated titles.

Conclusion

The development of the Takoyaki Mini House System demonstrates that even a small street-food business can benefit from enterprise-level software design. The project achieved its goal of creating a unified POS and Inventory platform that streamlines daily operations for both cashiers and management. The integration of real-time analytics provides the business with a competitive edge by enabling data-driven decisions on pricing, stock replenishment, and product performance. Role-based security ensures the system is safe from unauthorized access, protecting sensitive sales and inventory data. The most significant learning gain was the implementation of Fail-Safe JavaScript logic that handles complex JSON data without breaking the user interface, ensuring a stable and responsive experience even during high-transaction periods. Overall, the system proves that affordable, web-based solutions can modernize small food businesses and improve operational efficiency.

Recommendations

To further enhance the Takoyaki Mini House System, the following improvements are recommended:

- **Cloud Hosting:** Migrate from a local XAMPP environment to a cloud provider such as DigitalOcean or Heroku to enable remote management, improve accessibility, and ensure higher uptime for the business.
- **Sales Forecasting:** Leverage the collected transaction data to implement a basic AI-driven sales forecast for the next month, helping management anticipate demand, optimize stock levels, and reduce waste.
- **Integrated QR Ordering:** Allow customers to scan a QR code at the table to send their orders directly to the Cashier's cart, reducing wait times, minimizing manual entry errors, and improving the overall customer experience.
- **Multi-User Locking:** Implement a lock feature to prevent two cashiers from editing the same product at the exact same millisecond, ensuring data consistency and avoiding conflicts during high-traffic period.